Neural Computation -- 1989

53 article(s), all issues

1. Unsupervised Learning

H. B. Barlow

https://doi.org/10.1162/neco.1989.1.3.295 | pp. 1-18

What use can the brain make of the massive flow of sensory information that occurs without any associated rewards or punishments? This question is reviewed in the light of connectionist models of unsupervised learning and some older ideas, namely the cognitive maps and working models of Tolman and Craik, and the idea that redundancy is important for understanding perception (Attneave 1954), the physiology of sensory pathways (Barlow 1959), and pattern recognition (Watanabe 1960). It is argued that (1) The redundancy of sensory messages provides the knowledge incorporated in the maps or models. (2) Some of this knowledge can be obtained by observations of mean, variance, and covariance of sensory messages, and perhaps also by a method called “minimum entropy coding.” (3) Such knowledge may be incorporated in a model of “what usually happens” with which incoming messages are automatically compared, enabling unexpected discrepancies to be immediately identified. (4) Knowledge of the sort incorporated into such a filter is a necessary prerequisite of ordinary learning, and a representation whose elements are independent makes it possible to form associations with logical functions of the elements, not just with the elements themselves.

---

2. Review of Neural Networks for Speech Recognition

Richard P. Lippmann

https://doi.org/10.1162/neco.1989.1.1.1 | pp. 1-38

The performance of current speech recognition systems is far below that of humans. Neural nets offer the potential of providing massive parallelism, adaptation, and new algorithmic approaches to problems in speech recognition. Initial studies have demonstrated that multilayer networks with time delays can provide excellent discrimination between small sets of pre-segmented difficult-to-discriminate words, consonants, and vowels. Performance for these small vocabularies has often exceeded that of more conventional approaches. Physiological front ends have provided improved recognition accuracy in noise and a cochlea filter-bank that could be used in these front ends has been implemented using micro-power analog VLSI techniques. Techniques have been developed to scale networks up in size to handle larger vocabularies, to reduce training time, and to train nets with recurrent connections. Multilayer perceptron classifiers are being integrated into conventional continuous-speech recognizers. Neural net architectures have been developed to perform the computations required by vector quantizers, static pattern classifiers, and the Viterbi decoding algorithm. Further work is necessary for large-vocabulary continuous-speech problems, to develop training algorithms that progressively build internal word models, and to develop compact VLSI neural net hardware.

---

3. Modular Construction of Time-Delay Neural Networks for Speech Recognition

Alex Waibel

https://doi.org/10.1162/neco.1989.1.1.39 | pp. 39-46

Several strategies are described that overcome limitations of basic network models as steps towards the design of large connectionist speech recognition systems. The two major areas of concern are the problem of time and the problem of scaling. Speech signals continuously vary over time and encode and transmit enormous amounts of human knowledge. To decode these signals, neural networks must be able to use appropriate representations of time and it must be possible to extend these nets to almost arbitrary sizes and complexity within finite resources. The problem of time is addressed by the development of a Time-Delay Neural Network; the problem of scaling by Modularity and Incremental Design of large nets based on smaller subcomponent nets. It is shown that small networks trained to perform limited tasks develop time invariant, hidden abstractions that can subsequently be exploited to train larger, more complex nets efficiently. Using these techniques, phoneme recognition networks of increasing complexity can be constructed that all achieve superior recognition performance.

---

4. A Silicon Model Of Auditory Localization

John Lazzaro, Carver Mead

https://doi.org/10.1162/neco.1989.1.1.47 | pp. 47-57

The barn owl accurately localizes sounds in the azimuthal plane, using interaural time difference as a cue. The time-coding pathway in the owl's brainstem encodes a neural map of azimuth, by processing interaural timing information. We have built a silicon model of the time-coding pathway of the owl. The integrated circuit models the structure as well as the function of the pathway; most subcircuits in the chip have an anatomical correlate. The chip computes all outputs in real time, using analog, continuous-time processing.

---

5. Criteria for Robust Stability In A Class Of Lateral Inhibition Networks Coupled Through Resistive Grids

John L. Wyatt, David L. Standley

https://doi.org/10.1162/neco.1989.1.1.58 | pp. 58-67

In the analog VLSI implementation of neural systems, it is sometimes convenient to build lateral inhibition networks by using a locally connected on-chip resistive grid to interconnect active elements. A serious problem of unwanted spontaneous oscillation often arises with these circuits and renders them unusable in practice. This paper reports on criteria that guarantee these and certain other systems will be stable, even though the values of designed elements in the resistive grid may be imprecise and the location and values of parasitic elements may be unknown. The method is based on a rigorous, somewhat novel mathematical analysis using Tellegen's theorem (Penfield et al. 1970) from electrical circuits and the idea of a Popov multiplier (Vidyasagar 1978; Desoer and Vidya sagar 1975) from control theory. The criteria are local in that no overall analysis of the interconnected system is required for their use, empirical in that they involve only measurable frequency response data on the individual cells, and robust in that they are insensitive to network topology and to unmodelled parasitic resistances and capacitances in the interconnect network. Certain results are robust in the additional sense that specified nonlinear elements in the grid do not affect the stability criteria. The results are designed to be applicable, with further development, to complex and incompletely modeled living neural systems.

6. Two Stages of Curve Detection Suggest Two Styles of Visual Computation

Steven W. Zucker, A. Dobbins, Lee Iverson

https://doi.org/10.1162/neco.1989.1.1.68 | pp. 68-81

The problem of detecting curves in visual images arises in both computer vision and biological visual systems. Our approach integrates constraints from these two sources and suggests that there are two different stages to curve detection, the first resulting in a local description, and the second in a global one. Each stage involves a different style of computation: in the first stage, hypotheses are represented explicitly and coarsely in a fixed, preconfigured architecture; in the second stage, hypotheses are represented implicitly and more finely in a dynamically constructed architecture. We also show how these stages could be related to physiology, specifying the earlier parts in a relatively fine-grained fashion and the later ones more coarsely.

7. Part Segmentation for Object Recognition

Alex Pentland

https://doi.org/10.1162/neco.1989.1.1.82 | pp. 82-91

Visual object recognition is a difficult problem that has been solved by biological visual systems. An approach to object recognition is described in which the image is segmented into parts using two simple, biologically-plausible mechanisms: a filtering operation to produce a large set of potential object “parts,” followed by a new type of network that searches among these part hypotheses to produce the simplest, most likely description of the image's part structure.

8. Computing Optical Flow in the Primate Visual System

H. Taichi Wang, Bimal Mathur, Christof Koch

https://doi.org/10.1162/neco.1989.1.1.92 | pp. 92-103

Computing motion on the basis of the time-varying image intensity is a difficult problem for both artificial and biological vision systems. We show how gradient models, a well-known class of motion algorithms, can be implemented within the magnocellular pathway of the primate's visual system. Our cooperative algorithm computes optical flow in two steps. In the first stage, assumed to be located in primary visual cortex, local motion is measured while spatial integration occurs in the second stage, assumed to be located in the middle temporal area (MT). The final optical flow is extracted in this second stage using population coding, such that the velocity is represented by the vector sum of neurons coding for motion in different directions. Our theory, relating the single-cell to the perceptual level, accounts for a number of psychophysical and electrophysiological observations and illusions.

9. A Multiple-Map Model for Pattern Classification

Alan Rojer, Eric M. Schwartz

https://doi.org/10.1162/neco.1989.1.1.104 | pp. 104-115

A characteristic feature of vertebrate sensory cortex (and midbrain) is the existence of multiple two-dimensional map representations. Some workers have considered single-map classification (e.g. Kohonen 1984) but little work has focused on the use of multiple maps. We have constructed a multiple-map classifier, which permits abstraction of the computational properties of a multiple-map architecture. We identify three problems which characterize a multiple-map classifier: classification in two dimensions, mapping from high dimensions to two dimensions, and combination of multiple maps. We demonstrate component solutions to each of the problems, using Parzen-window density estimation in two dimensions, a generalized Fisher discriminant function for dimensionality reduction, and split/merge methods to construct a “tree of maps” for the multiple-map representation. The combination of components is modular and each component could be improved or replaced without affecting the other components. The classifier training procedure requires time linear in the number of training examples; classification time is independent of the number of training examples and requires constant space. Performance of this classifier on Fisher's iris data, Gaussian clusters on a five-dimensional simplex, and digitized speech data is comparable to competing algorithms, such as nearest-neighbor, back-propagation and Gaussian classifiers. This work provides an example of the computational utility of multiple-map representations for classification. It is one step towards the goal of understanding why brain areas such as visual cortex utilize multiple map-like representations of the world.

---

10. A Control Systems Model of Smooth Pursuit Eye Movements with Realistic Emergent Properties

Richard J. Krauzlis, Stephen G. Lisberger

https://doi.org/10.1162/neco.1989.1.1.116 | pp. 116-122

Visual tracking of objects in a noisy environment is a difficult problem that has been solved by the primate oculomotor system, but remains unsolved in robotics. In primates, smooth pursuit eye movements match eye motion to target motion to keep the eye pointed at smoothly moving targets. We have used computer models as a tool to investigate possible computational strategies underlying this behavior. Here, we present a model based upon behavioral data from monkeys. The model emphasizes the variety of visual signals available for pursuit and, in particular, includes a sensitivity to the acceleration of retinal images. The model was designed to replicate the initial eye velocity response observed during pursuit of different target motions. The strength of the model is that it also exhibits a number of emergent properties that are seen in the behavior of both humans and monkeys. This suggests that the elements in the model capture important aspects of the mechanism of visual tracking by the primate smooth pursuit system.

---

11. The Brain Binds Entities and Events by Multiregional Activation from Convergence Zones

António R. Damásio

https://doi.org/10.1162/neco.1989.1.1.123 | pp. 123-132

The experience of reality, in both perception and recall, is spatially and temporally coherent and “in-register.” Features are bound in entities, and entities are bound in events. The properties of these entities and events, however, are represented in many different regions of the brain that are widely separated. The degree of neural parcellation is even greater when we consider that the perception of most entities and events also

requires a motor interaction on the part of the perceiver (such as eye movements and hand movements) and often includes a recordable modification of the perceiver's somatic state. The question of how the brain achieves integration starting with the bits and pieces it has to work with, is the binding problem. Here we propose a new solution for this problem, at the level of neural systems that integrate functional regions of the telencephalon.

---

12. Product Units: A Computationally Powerful and Biologically Plausible Extension to Backpropagation Networks

Richard Durbin, David E. Rumelhart

https://doi.org/10.1162/neco.1989.1.1.133 | pp. 133-142

We introduce a new form of computational unit for feedforward learning networks of the backpropagation type. Instead of calculating a weighted sum this unit calculates a weighted product, where each input is raised to a power determined by a variable weight. Such a unit can learn an arbitrary polynomial term, which would then feed into higher level standard summing units. We show how learning operates with product units, provide examples to show their efficiency for various types of problems, and argue that they naturally extend the family of theoretical feedforward net structures. There is a plausible neurobiological interpretation for one interesting configuration of product and summing units.

---

13. Deterministic Boltzmann Learning Performs Steepest Descent in Weight-Space

Geoffrey E. Hinton

https://doi.org/10.1162/neco.1989.1.1.143 | pp. 143-150

The Boltzmann machine learning procedure has been successfully applied in deterministic networks of analog units that use a mean field approximation to efficiently simulate a truly stochastic system (Peterson and Anderson 1987). This type of “deterministic Boltzmann machine” (DBM) learns much faster than the equivalent “stochastic Boltzmann machine” (SBM), but since the learning procedure for DBM's is only based on an analogy with SBM's, there is no existing proof that it performs gradient descent in any function, and it has only been justified by simulations. By using the appropriate interpretation for the way in which a DBM represents the probability of an output vector given an input vector, it is shown that the DBM performs steepest descent in the same function as the original SBM, except at rare discontinuities. A very simple way of forcing the weights to become symmetrical is also described, and this makes the DBM more biologically plausible than back-propagation (Werbos 1974; Parker 1985; Rumelhart et al. 1986).

---

14. What Size Net Gives Valid Generalization?

Eric B. Baum, David Haussler

https://doi.org/10.1162/neco.1989.1.1.151 | pp. 151-160

We address the question of when a network can be expected to generalize from m random training examples chosen from some arbitrary probability distribution, assuming that future test examples are drawn from the same distribution. Among our results are the following bounds on appropriate sample vs. network

size. Assume 0 < ■ ≤ 1/8. We show that if m ≥ O(W/■ log N/■) random examples can be loaded on a feedforward network of linear threshold functions with N nodes and W weights, so that at least a fraction 1 – ■/2 of the examples are correctly classified, then one has confidence approaching certainty that the network will correctly classify a fraction 1 – ■ of future test examples drawn from the same distribution. Conversely, for fully-connected feedforward nets with one hidden layer, any learning algorithm using fewer than Ω(W/■) random training examples will, for some distributions of examples consistent with an appropriate weight choice, fail at least some fixed fraction of the time to find a weight choice that will correctly classify more than a 1 – ■ fraction of the future test examples.

---

15. Recurrent Backpropagation and the Dynamical Approach to Adaptive Neural Computation

Fernando J. Pineda

https://doi.org/10.1162/neco.1989.1.2.161 | pp. 161-172

Error backpropagation in feedforward neural network models is a popular learning algorithm that has its roots in nonlinear estimation and optimization. It is being used routinely to calculate error gradients in nonlinear systems with hundreds of thousands of parameters. However, the classical architecture for backpropagation has severe restrictions. The extension of backpropagation to networks with recurrent connections will be reviewed. It is now possible to efficiently compute the error gradients for networks that have temporal dynamics, which opens applications to a host of problems in systems identification and control.

---

16. New Models for Motor Control

Jennifer S. Altman, Jenny Kien

https://doi.org/10.1162/neco.1989.1.2.173 | pp. 173-183

What do a prototype robot (Brooks 1989) and a model for the control of behavioral choice in insects (Altman and Kien 1987a) have in common? And what do they share with a scratching cat (Shadmehr 1989)? The answer is distributed control systems that do not depend on a central command center for the execution of behavioral outputs. The first two in particular are examples of a growing trend to replace the long-held concept of linear hierarchical control of motor output with one of decentralized, distributed control, with inputs at many levels and the output a consensus of the activity in several centers. Brooks (1989) describes a six-legged machine that, in its most advanced form, can walk over rough terrain and prowl around following a source of warmth, such as a person. The six legs, chosen as a compromise between stability and ease of coordination, give the robot a superficial resemblance to an insect — but the similarity goes deeper. The modular control system, designed strictly on engineering principles for maximum efficiency and economy, bears a striking similarity to the model we have proposed elsewhere (Altman and Kien 1987a) to describe the organization of the motor system in insects such as the locust. In both systems, the same set of components can generate different behaviors, depending on the context, and similar principles govern the generation of different levels of behavior, from movements of a single leg to coordinated responses of the whole beast. Neither requires a single center for integrating all sensory information and conflicts tend to be resolved by consensus at the motor level.

---

17. Seeing Chips: Analog VLSI Circuits for Computer Vision

Christof Koch

https://doi.org/10.1162/neco.1989.1.2.184 | pp. 184-200

Vision is simple. We open our eyes and, instantly, the world surrounding us is perceived in all its splendor. Yet Artificial Intelligence has been trying with very limited success for over 20 years to endow machines with similar abilities. A large van, filled with computers and driving unguided at a mile per hour across gently sloping hills in Colorado and using a laser-range system to “see” is the most we have accomplished so far. On the other hand, computers can play a decent game of chess or prove simple mathematical theorems. It is ironic that we are unable to reproduce perceptual abilities which we share with most animals while some of the features distinguishing us from even our closest cousins, chimpanzees, can be carried out by machines. Vision is difficult.

---

18. A Proposal for More Powerful Learning Algorithms

Eric B. Baum

https://doi.org/10.1162/neco.1989.1.2.201 | pp. 201-207

Judd (1988) and Blum and Rivest (1988) have recently proved that the loading problem for neural networks is NP complete. This makes it very unlikely that any algorithm like backpropagation which varies weights on a network of fixed size and topology will be able to learn in polynomial time. However, Valiant has recently proposed a learning protocol (Valiant 1984), which allows one to sensibly consider generalization by learning algorithms with the freedom to add neurons and synapses, as well as simply adjusting weights. Within this context, standard circuit complexity arguments show that learning algorithms with such freedom can solve in polynomial time any learning problem that can be solved in polynomial time by any algorithm whatever. In this sense, neural nets are universal learners, capable of learning any learnable class of concepts.

---

19. A Possible Neural Mechanism for Computing Shape From Shading

Alex Pentland

https://doi.org/10.1162/neco.1989.1.2.208 | pp. 208-217

A simple neural mechanism that recovers surface shape from image shading is derived from a simplified model of the physics of image formation. The mechanism's performance is surprisingly good even when applied to complex natural images, and is even able to extract significant shape information from some line drawings.

---

20. Optimization in Model Matching and Perceptual Organization

Eric Mjolsness, Gene Gindi, P. Anandan

https://doi.org/10.1162/neco.1989.1.2.218 | pp. 218-229

We introduce an optimization approach for solving problems in computer vision that involve multiple levels of abstraction. Our objective functions include compositional and specialization hierarchies. We cast vision

problems as inexact graph matching problems, formulate graph matching in terms of constrained optimization, and use analog neural networks to perform the optimization. The method is applicable to perceptual grouping and model matching. Preliminary experimental results are shown.

---

21. Distributed Parallel Processing in the Vestibulo-Oculomotor System

Thomas J. Anastasio, David A. Robinson

https://doi.org/10.1162/neco.1989.1.2.230 | pp. 230-241

The mechanisms of eye-movement control are among the best understood in motor neurophysiology. Detailed anatomical and physiological data have paved the way for theoretical models that have unified existing knowledge and suggested further experiments. These models have generally taken the form of black-box diagrams (for example, Robinson 1981) representing the flow of hypothetical signals between idealized signal-processing blocks. They approximate overall oculomotor behavior but indicate little about how real eye-movement signals would be carried and processed by real neural networks. Neurons that combine and transmit oculomotor signals, such as those in the vestibular nucleus (VN), actually do so in a diverse, seemingly random way that would be impossible to predict from a block diagram. The purpose of this study is to use a neural-network learning scheme (Rumelhart et al. 1986) to construct parallel, distributed models of the vestibulo-oculomotor system that simulate the diversity of responses recorded experimentally from VN neurons.

---

22. A Neural Model for Generation of Some Behaviors in the Fictive Scratch Reflex

Reza Shadmehr

https://doi.org/10.1162/neco.1989.1.2.242 | pp. 242-252

We have studied the scratch reflex in order to better understand the strategy that the spinal neural structures use to program limb movements. A network is proposed for the positioning and rhythmic portions of the scratch reflex in the deafferented preparation. This model is based on Berkinblit's hypotheses regarding organization of neurons constituting the spinal central pattern generator (CPG) for this behavior. We offer a mechanism by which the initial position of the hindlimb can influence the output of the CPG in the immobilized animal.

---

23. A Robot that Walks; Emergent Behaviors from a Carefully Evolved Network

Rodney A. Brooks

https://doi.org/10.1162/neco.1989.1.2.253 | pp. 253-262

Most animals have significant behavioral expertise built in without having to explicitly learn it all from scratch. This expertise is a product of evolution of the organism; it can be viewed as a very long-term form of learning which provides a structured system within which individuals might learn more specialized skills or abilities. This paper suggests one possible mechanism for analagous robot evolution by describing a carefully designed series of networks, each one being a strict augmentation of the previous one, which control a

six-legged walking machine capable of walking over rough terrain and following a person passively sensed in the infrared spectrum. As the completely decentralized networks are augmented, the robot's performance and behavior repertoire demonstrably improve. The rationale for such demonstrations is that they may provide a hint as to the requirements for automatically building massive networks to carry out complex sensory-motor tasks. The experiments with an actual robot ensure that an essence of reality is maintained and that no critical disabling problems have been ignored.

---

24. Learning State Space Trajectories in Recurrent Neural Networks

Barak A. Pearlmutter

https://doi.org/10.1162/neco.1989.1.2.263 | pp. 263-269

Many neural network learning procedures compute gradients of the errors on the output layer of units after they have settled to their final values. We describe a procedure for finding ∂E/∂w ij , where E is an error functional of the temporal trajectory of the states of a continuous recurrent network and w ij are the weights of that network. Computing these quantities allows one to perform gradient descent in the weights to minimize E. Simulations in which networks are taught to move through limit cycles are shown. This type of recurrent network seems particularly suited for temporally continuous domains, such as signal processing, control, and speech.

---

25. A Learning Algorithm for Continually Running Fully Recurrent Neural Networks

Ronald J. Williams, David Zipser

https://doi.org/10.1162/neco.1989.1.2.270 | pp. 270-280

The exact form of a gradient-following learning algorithm for completely recurrent networks running in continually sampled time is derived and used as the basis for practical algorithms for temporal supervised learning tasks. These algorithms have (1) the advantage that they do not require a precisely defined training interval, operating while the network runs; and (2) the disadvantage that they require nonlocal communication in the network being trained and are computationally expensive. These algorithms allow networks having recurrent connections to learn complex tasks that require the retention of information over time periods having either fixed or indefinite length.

---

26. Fast Learning in Networks of Locally-Tuned Processing Units

John Moody, Christian J. Darken

https://doi.org/10.1162/neco.1989.1.2.281 | pp. 281-294

We propose a network architecture which uses a single internal layer of locally-tuned processing units to learn both classification tasks and real-valued function approximations (Moody and Darken 1988). We consider training such networks in a completely supervised manner, but abandon this approach in favor of a more computationally efficient hybrid learning method which combines self-organized and supervised learning. Our networks learn faster than backpropagation for two reasons: the local representations ensure that only a few units respond to any given input, thus reducing computational overhead, and the hybrid

learning rules are linear rather than nonlinear, thus leading to faster convergence. Unlike many existing methods for data analysis, our network architecture and learning rules are truly adaptive and are thus appropriate for real-time use.

---

27. The Vapnik-Chervonenkis Dimension: Information versus Complexity in Learning

Yaser S. Abu■Mostafa

https://doi.org/10.1162/neco.1989.1.3.312 | pp. 312-317

When feasible, learning is a very attractive alternative to explicit programming. This is particularly true in areas where the problems do not lend themselves to systematic programming, such as pattern recognition in natural environments. The feasibility of learning an unknown function from examples depends on two questions: 1. Do the examples convey enough information to determine the function? 2. Is there a speedy way of constructing the function from the examples? These questions contrast the roles of information and complexity in learning. While the two roles share some ground, they are conceptually and technically different. In the common language of learning, the information question is that of generalization and the complexity question is that of scaling. The work of Vapnik and Chervonenkis (1971) provides the key tools for dealing with the information issue. In this review, we develop the main ideas of this framework and discuss how complexity fits in.

---

28. Linking Linear Threshold Units with Quadratic Models of Motion Perception

Humbert H. Suarez, Christof Koch

https://doi.org/10.1162/neco.1989.1.3.318 | pp. 318-320

Behavioral experiments on insects (Hassenstein and Reichardt 1956; Poggio and Reichardt 1976) as well as psychophysical evidence from human studies (Van Santen and Sperling 1985; Adelson and Bergen 1985; Watson and Ahumada 1985) support the notion that short-range motion perception is mediated by a system with a quadratic type of nonlinearity, as in correlation (Hassenstein and Reichardt 1956), multiplication (Torre and Poggio 1978), or squaring (Adelson and Bergen 1985). However, there is little physiological evidence for quadratic nonlinearities in directionally selective cells. For instance, the response of cortical simple cells to a moving sine grating is half-wave instead of full-wave rectified as it should be for a quadratic nonlinearity (Movshon ef al. 1978; Holub and Morton-Gibson 1981) and is linear for low contrast (Holub and Morton-Gibson 1981). Complex cells have full-wave rectified responses, but are also linear in contrast. Moreover, a detailed theoretical analysis of possible biophysical mechanisms underlying direction selectivity concludes that most do not have quadratic properties except under very limited conditions (Grzywacz and Koch 1987). Thus, it is presently mysterious how a system can show quadratic properties while its individual components do not. We briefly discuss here a simple population encoding scheme offering a possible solution to this problem.

---

29. Neurogammon Wins Computer Olympiad

Gerald Tesauro
https://doi.org/10.1162/neco.1989.1.3.321 | pp. 321-323

Neurogammon 1.0 is a backgammon program which uses multilayer neural networks to make move decisions and doubling decisions. The networks learned to play backgammon by backpropagation training on expert data sets. At the recently held First Computer Olympiad in London, Neurogammon won the backgammon competition with a perfect record of five wins and no losses, thereby becoming the first learning program ever to win a tournament.

---

30. Surface Interpolation in Three-Dimensional Structure-from-Motion Perception

Masud Husain, Stefan Treue, Richard A. Andersen
https://doi.org/10.1162/neco.1989.1.3.324 | pp. 324-333

Although it is appreciated that humans can use a number of visual cues to perceive the three-dimensional (3-D) shape of an object, for example, luminance, orientation, binocular disparity, and motion, the exact mechanisms employed are not known (De Yoe and Van Essen 1988). An important approach to understanding the computations performed by the visual system is to develop algorithms (Marr 1982) or neural network models (Lehky and Sejnowski 1988; Siegel 1987) that are capable of computing shape from specific cues in the visual image. In this study we investigated the ability of observers to see the 3-D shape of an object using motion cues, so called structure-from-motion (SFM). We measured human performance in a two-alternative forced choice task using novel dynamic random-dot stimuli with limited point lifetimes. We show that the human visual system integrates motion information spatially and temporally (across several point lifetimes) as part of the process for computing SFM. We conclude that SFM algorithms must include surface interpolation to account for human performance. Our experiments also provide evidence that local velocity information, and not position information derived from discrete views of the image (as proposed by some algorithms), is used to solve the SFM problem by the human visual system.

---

31. A Winner-Take-All Mechanism Based on Presynaptic Inhibition Feedback

Alan Yuille, Norberto M. Grzywacz
https://doi.org/10.1162/neco.1989.1.3.334 | pp. 334-347

A winner-take-all mechanism is a device that determines the identity and amplitude of its largest input (Feldman and Ballard 1982). Such mechanisms have been proposed for various brain functions. For example, a theory for visual velocity estimate (Grzywacz and Yuille 1989) postulates that a winner-take-all selects the strongest responding cell in the cortex's middle temporal area (MT). This theory proposes a circuitry that links the directionally selective cells in the primary visual cortex to MT cells, making them velocity selective. Generally, several velocity cells would respond, but only the winner would determine the perception. In another theory, a winner-take-all guides the spotlight of attention to the most salient image part (Koch and Ullman 1985). Also, such mechanisms improve the signal-to-noise ratios of VLSI emulations of brain functions (Lazzaro and Mead 1989). Although computer algorithms for winner-take-all mechanisms exist (Feldman and Ballard 1982; Koch and Ullman 1985), good biologically motivated models do not. A candidate for a biological mechanism is lateral (mutual) inhibition (Hartline and Ratliff 1957). In some

theoretical mutual-inhibition networks, the inhibition sums linearly to the excitatory inputs and the result is passed through a threshold non linearity (Hadeler 1974). However, these networks work only if the difference between winner and losers is large (Koch and Ullman 1985). We propose an alternative network, in which the output of each element feeds back to inhibit the inputs to other elements. The action of this presynaptic inhibition is nonlinear with a possible biophysical substrate. This paper shows that the new network converges stably to a solution that both relays the winner's identity and amplitude and suppresses information on the losers with arbitrary precision. We prove these results mathematically and illustrate the effectiveness of the network and some of its variants by computer simulations.

---

32. An Analysis of the Elastic Net Approach to the Traveling Salesman Problem

Richard Durbin, Richard Szeliski, Alan Yuille

https://doi.org/10.1162/neco.1989.1.3.348 | pp. 348-358

This paper analyzes the elastic net approach (Durbin and Willshaw 1987) to the traveling salesman problem of finding the shortest path through a set of cities. The elastic net approach jointly minimizes the length of an arbitrary path in the plane and the distance between the path points and the cities. The tradeoff between these two requirements is controlled by a scale parameter K. A global minimum is found for large K, and is then tracked to a small value. In this paper, we show that (1) in the small K limit the elastic path passes arbitrarily close to all the cities, but that only one path point is attracted to each city, (2) in the large K limit the net lies at the center of the set of cities, and (3) at a critical value of K the energy function bifurcates. We also show that this method can be interpreted in terms of extremizing a probability distribution controlled by K. The minimum at a given K corresponds to the maximum a posteriori (MAP) Bayesian estimate of the tour under a natural statistical interpretation. The analysis presented in this paper gives us a better understanding of the behavior of the elastic net, allows us to better choose the parameters for the optimization, and suggests how to extend the underlying ideas to other domains.

---

33. The Storage of Time Intervals Using Oscillating Neurons

Christopher Miall

https://doi.org/10.1162/neco.1989.1.3.359 | pp. 359-371

A mechanism to store and recall time intervals ranging from hundreds of milliseconds to tens of seconds is described. The principle is based on beat frequencies between oscillating elements; any small group of oscillators codes specifically for an interval equal to the lowest common multiple of their oscillation periods. This mechanism could be realized in the nervous system by an output neuron, excited by a group of pacemaker neurons, and able to select via a Hebbian rule a subgroup of pacemaker cells to encode any given interval, or small number of intervals (for example, a pattern of pulses). Recall could be achieved by resetting the pacemaker cells and setting a threshold for activation of the output unit. A simulation is described and the main features of such an encoding scheme are discussed.

---

34. Finite State Automata and Simple Recurrent Networks

Axel Cleeremans, David Servan■Schreiber, James L. McClelland
https://doi.org/10.1162/neco.1989.1.3.372 | pp. 372-381

We explore a network architecture introduced by Elman (1988) for predicting successive elements of a sequence. The network uses the pattern of activation over a set of hidden units from time-step t–1, together with element t, to predict element t + 1. When the network is trained with strings from a particular finite-state grammar, it can learn to be a perfect finite-state recognizer for the grammar. When the network has a minimal number of hidden units, patterns on the hidden units come to correspond to the nodes of the grammar, although this correspondence is not necessary for the network to act as a perfect finite-state recognizer. We explore the conditions under which the network can carry information about distant sequential contingencies across intervening elements. Such information is maintained with relative ease if it is relevant at each intermediate step; it tends to be lost when intervening elements do not depend on it. At first glance this may suggest that such networks are not relevant to natural language, in which dependencies may span indefinite distances. However, embeddings in natural language are not completely independent of earlier information. The final simulation shows that long distance sequential contingencies can be encoded by the network even if only subtle statistical properties of embedded strings depend on the early information.

---

35. Asymptotic Convergence of Backpropagation

Gerald Tesauro, Yu He, Subutai Ahmad
https://doi.org/10.1162/neco.1989.1.3.382 | pp. 382-391

We calculate analytically the rate of convergence at long times in the backpropagation learning algorithm for networks with and without hidden units. For networks without hidden units using the standard quadratic error function and a sigmoidal transfer function, we find that the error decreases as 1/t for large t, and the output states approach their target values as $1/\sqrt{t}$. It is possible to obtain a different convergence rate for certain error and transfer functions, but the convergence can never be faster than 1/t. These results are unaffected by a momentum term in the learning algorithm, but convergence can be substantially improved by an adaptive learning rate scheme. For networks with hidden units, we generally expect the same rate of convergence to be obtained as in the single-layer case; however, under certain circumstances one can obtain a polynomial speed-up for non sigmoidal units, or a logarithmic speed-up for sigmoidal units. Our analytic results are confirmed by empirical measurements of the convergence rate in numerical simulations.

---

36. Learning by assertion: A method for calibrating a simple visual system

Laurence T. Maloney, A. J. Ahumada
(no DOI) | pp. 387-395

No abstract available.

---

37. Learning by Assertion: Two Methods for Calibrating a Linear Visual System

Laurence T. Maloney, Albert J. Ahumada
https://doi.org/10.1162/neco.1989.1.3.392 | pp. 392-401

A visual system is geometrically calibrated if its estimates of the spatial properties of a scene are accurate: straight lines are judged straight, angles are correctly estimated, and collinear line segments are perceived to fall on a common line. This paper describes two new calibration methods for a model visual system whose photoreceptors are initially at unknown locations. The methods can also compensate for optical distortions that are equivalent to remapping of receptor locations (e.g., spherical aberration). The methods work by comparing visual input across eye/head movements; they require no explicit feedback and no knowledge about the particular contents of a scene. This work has implications for development and calibration in biological visual systems.

---

38. How to Generate Ordered Maps by Maximizing the Mutual Information between Input and Output Signals

Ralph Linsker

https://doi.org/10.1162/neco.1989.1.3.402 | pp. 402-411

A learning rule that performs gradient ascent in the average mutual information between input and an output signal is derived for a system having feedforward and lateral interactions. Several processes emerge as components of this learning rule: Hebb-like modification, and cooperation and competition among processing nodes. Topographic map formation is demonstrated using the learning rule. An analytic expression relating the average mutual information to the response properties of nodes and their geometric arrangement is derived in certain cases. This yields a relation between the local map magnification factor and the probability distribution in the input space. The results provide new links between unsupervised learning and information-theoretic optimization in a system whose properties are biologically motivated.

---

39. Finding Minimum Entropy Codes

H. B. Barlow, T.P. Kaushal, Graeme Mitchison

https://doi.org/10.1162/neco.1989.1.3.412 | pp. 412-423

To determine whether a particular sensory event is a reliable predictor of reward or punishment it is necessary to know the prior probability of that event. If the variables of a sensory representation normally occur independently of each other, then it is possible to derive the prior probability of any logical function of the variables from the prior probabilities of the individual variables, without any additional knowledge; hence such a representation enormously enlarges the scope of definable events that can be searched for reliable predictors. Finding a Minimum Entropy Code is a possible method of forming such a representation, and methods for doing this are explored in this paper. The main results are (1) to show how to find such a code when the probabilities of the input states form a geometric progression, as is shown to be nearly true for keyboard characters in normal text; (2) to show how a Minimum Entropy Code can be approximated by repeatedly recoding pairs, triples, etc. of an original 7-bit code for keyboard characters; (3) to prove that in some cases enlarging the capacity of the output channel can lower the entropy.

---

40. Learning in Artificial Neural Networks: A Statistical Perspective

Halbert White

https://doi.org/10.1162/neco.1989.1.4.425 | pp. 425-464

The premise of this article is that learning procedures used to train artificial neural networks are inherently statistical techniques. It follows that statistical theory can provide considerable insight into the properties, advantages, and disadvantages of different network learning methods. We review concepts and analytical results from the literatures of mathematical statistics, econometrics, systems identification, and optimization theory relevant to the analysis of learning in artificial neural networks. Because of the considerable variety of available learning procedures and necessary limitations of space, we cannot provide a comprehensive treatment. Our focus is primarily on learning procedures for feedforward networks. However, many of the concepts and issues arising in this framework are also quite broadly relevant to other network learning paradigms. In addition to providing useful insights, the material reviewed here suggests some potentially useful new training methods for artificial neural networks.

---

41. Representation Properties of Networks: Kolmogorov's Theorem Is Irrelevant

Federico Girosi, Tomaso Poggio

https://doi.org/10.1162/neco.1989.1.4.465 | pp. 465-469

Many neural networks can be regarded as attempting to approximate a multivariate function in terms of one-input one-output units. This note considers the problem of an exact representation of nonlinear mappings in terms of simpler functions of fewer variables. We review Kolmogorov's theorem on the representation of functions of several variables in terms of functions of one variable and show that it is irrelevant in the context of networks for learning.

---

42. Sigmoids Distinguish More Efficiently Than Heavisides

Eduardo D. Sontag

https://doi.org/10.1162/neco.1989.1.4.470 | pp. 470-472

Every dichotomy on a 2k-point set in ■ N can be implemented by a neural net with a single hidden layer containing k sigmoidal neurons. If the neurons were of a hardlimiter (Heaviside) type, 2k – 1 would be in general needed.

---

43. How Cortical Interconnectedness Varies with Network Size

Charles F. Stevens

https://doi.org/10.1162/neco.1989.1.4.473 | pp. 473-479

When model neural networks are used to gain insight into how the brain might carry out its computations, comparisons between features of the network and those of the brain form an important basis for drawing conclusions about the network's relevance to brain function. The most significant features to be compared, of course, relate to behavior of the units. Another network property that would be useful to consider, however, is the extent to which units are interconnected and the law by which unit-unit connections scale as the network is made larger. The goal of this paper is to consider these questions for neocortex. The conclusion will be that neocortical neurons are rather sparsely interconnected — each neuron receives

direct synaptic input from fewer than 3% of its neighbors underlying the surrounding square millimeter of cortex — and the extent of connectedness hardly changes for brains that range in size over about four orders of magnitude. These conclusions support the currently popular notion that the brain's circuits are highly modular and suggest that increased cortex size is mainly achieved by adding more modules.

---

44. A Canonical Microcircuit for Neocortex

Rodney J. Douglas, Kevan A Martin, D. Whitteridge

https://doi.org/10.1162/neco.1989.1.4.480 | pp. 480-488

We have used microanatomy derived from single neurons, and in vivo intracellular recordings to develop a simplified circuit of the visual cortex. The circuit explains the intracellular responses to pulse stimulation in terms of the interactions between three basic populations of neurons, and reveals the following features of cortical processing that are important to computational theories of neocortex. First, inhibition and excitation are not separable events. Activation of the cortex inevitably sets in motion a sequence of excitation and inhibition in every neuron. Second, the thalamic input does not provide the major excitation arriving at any neuron. Instead the intracortical excitatory connections provide most of the excitation. Third, the time evolution of excitation and inhibition is far longer than the synaptic delays of the circuits involved. This means that cortical processing cannot rely on precise timing between individual synaptic inputs.

---

45. Synthetic Neural Circuits Using Current-Domain Signal Representations

Andreas G. Andreou, Kwabena Boahen

https://doi.org/10.1162/neco.1989.1.4.489 | pp. 489-501

We present a new approach to the engineering of collective analog computing systems that emphasizes the role of currents as an appropriate signal representation and the need for low-power dissipation and simplicity in the basic functional circuits. The design methodology and implementation style that we describe are inspired by the functional and organizational principles of neuronal circuits in living systems. We have implemented synthetic neurons and synapses in analog CMOS VLSI that are suitable for building associative memories and self-organizing feature maps.

---

46. Random Neural Networks with Negative and Positive Signals and Product Form Solution

Erol Gelenbe

https://doi.org/10.1162/neco.1989.1.4.502 | pp. 502-510

We introduce a new class of random “neural” networks in which signals are either negative or positive. A positive signal arriving at a neuron increases its total signal count or potential by one; a negative signal reduces it by one if the potential is positive, and has no effect if it is zero. When its potential is positive, a neuron “fires,” sending positive or negative signals at random intervals to neurons or to the outside. Positive signals represent excitatory signals and negative signals represent inhibition. We show that this model, with exponential signal emission intervals, Poisson external signal arrivals, and Markovian signal movements between neurons, has a product form leading to simple analytical expressions for the system state.

47. Nonlinear Optimization Using Generalized Hopfield Networks

Athanasios G. Tsirukis, Gintaras V. Reklaitis, M.F. Tenorio

https://doi.org/10.1162/neco.1989.1.4.511 | pp. 511-521

A nonlinear neural framework, called the generalized Hopfield network (GHN), is proposed, which is able to solve in a parallel distributed manner systems of nonlinear equations. The method is applied to the general nonlinear optimization problem. We demonstrate GHNs implementing the three most important optimization algorithms, namely the augmented Lagrangian, generalized reduced gradient, and successive quadratic programming methods. The study results in a dynamic view of the optimization problem and offers a straightforward model for the parallelization of the optimization computations, thus significantly extending the practical limits of problems that can be formulated as an optimization problem and that can gain from the introduction of nonlinearities in their structure (e.g., pattern recognition, supervised learning, and design of content-addressable memories).

48. Discrete Synchronous Neural Algorithm for Minimization

Hyuk Lee

https://doi.org/10.1162/neco.1989.1.4.522 | pp. 522-531

A general discrete minimization algorithm that can be implemented by highly parallel neural networks is developed. It can be applied to the energy functions that can be expressed as arbitrary types of polynomial functions of the state variables. The algorithm can be operated in a synchronous way.

49. Approximation of Boolean Functions by Sigmoidal Networks: Part I: XOR and Other Two-Variable Functions

E. K. Blum

https://doi.org/10.1162/neco.1989.1.4.532 | pp. 532-540

We prove the existence of a manifold of exact solutions (mean-square error E = 0) of weights and thresholds for sigmoidal networks for XOR and other 2-variable Boolean functions. We also prove the existence of a manifold of local minima of E where $E \neq 0$.

50. Backpropagation Applied to Handwritten Zip Code Recognition

Yann LeCun, Bernhard E. Boser, J. S. Denker, D. Henderson, Richard Howard, W. Hubbard, L. D. Jackel

https://doi.org/10.1162/neco.1989.1.4.541 | pp. 541-551

The ability of learning networks to generalize can be greatly enhanced by providing constraints from the task domain. This paper demonstrates how such constraints can be integrated into a backpropagation network through the architecture of the network. This approach has been successfully applied to the recognition of handwritten zip code digits provided by the U.S. Postal Service. A single network learns the entire recognition operation, going from the normalized image of the character to the final classification.

51. A Subgrouping Strategy that Reduces Complexity and Speeds Up Learning in Recurrent Networks

David Zipser

https://doi.org/10.1162/neco.1989.1.4.552 | pp. 552-558

An algorithm, called RTRL, for training fully recurrent neural networks has recently been studied by Williams and Zipser (1989a, b). Whereas RTRL has been shown to have great power and generality, it has the disadvantage of requiring a great deal of computation time. A technique is described here for reducing the amount of computation required by RTRL without changing the connectivity of the networks. This is accomplished by dividing the original network into subnets for the purpose of error propagation while leaving them undivided for activity propagation. An example is given of a 12-unit network that learns to be the finite-state part of a Turing machine and runs 10 times faster using the subgrouping strategy than the original algorithm.

---

52. Unification as Constraint Satisfaction in Structured Connectionist Networks

Andreas Stolcke

https://doi.org/10.1162/neco.1989.1.4.559 | pp. 559-567

Unification is a basic concept in several traditional symbolic formalisms that should be well suited for a connectionist implementation due to the intuitive nature of the notions it formalizes. It is shown that by approaching unification from a graph matching and constraint satisfaction perspective a natural and efficient realization in a structured connectionist network can be found.

---

53. Index to Volume 1

(authors unknown)

https://doi.org/10.1162/neco.1989.1.4.568 | pp. 568-573

No abstract available.